

# Clinical Risk Assessment Report: Patient 61

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 39.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.78x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM\_LGZE.CT), which elevated the patient's absolute risk by +7.2%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Large Zone Emphasis (CT) (GLZLM\_LGZE.CT)** **Value: -0.35 [Bottom 12%] (+7.2%)**  
*Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.*

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)** **Value: 0.77 [Top 22%] (+3.7%)**  
*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

**Squamous Cell Carcinoma Antigen (SCC\_Ag)** **Value: 0.6 [Bottom 18%] (+3.5%)**  
*Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.*

**Maximum Tumor Density (CONVENTIONAL\_HUmax)** **Value: 0.71 [Top 16%] (+3.0%)**  
*Reflects the most dense solid components or calcifications within the primary tumor lesion.*

**Short Run High Gray-Level Emphasis (CT) (GLRLM\_SRHGE.CT)** **Value: 0.86 [Top 21%] (+2.7%)**  
*Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)** **Value: 0.71 [Avg (70th %ile)] (+1.7%)**  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Short Run Low Gray-Level Emphasis (CT) (GLRLM\_SRLGE.CT)** **Value: -0.19 [Avg (26th %ile)] (+1.5%)**  
*Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.*

## 3. Protective / Mitigating Factors (Reducing Risk)

**Tumor Sphericity (CT) (SHAPE\_Sphericity(onlyFor3DROI)).CT)** **Value: 0.96 [Top 16%] (-3.4%)**  
*Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.*

# Machine Learning Feature Attribution (SHAP Decision Path)

